

DW_sra

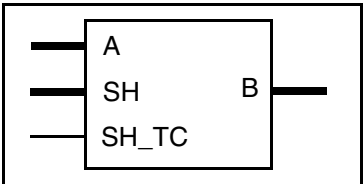
Arithmetic Shifter with Preferred Right Direction (VHDL style)

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Features and Benefits

- Parameterized data and shift coefficient word lengths
- Uses VHDL semantics for the arithmetic shift operation
- Capable of shifting in both directions

Revision History



Description

DW_sra is an arithmetic shifter that has the same semantics as the sra operator in VHDL. A list of input and output pins is shown in [Table 1-1](#). The component has the parameters shown in [Table 1-2](#). Parameters control the number of bits used in the component's ports.

The DW_sra may be configured to work as a bidirectional or unidirectional shifter. The SH_TC input indicates if the shifting distance (SH) is positive or negative. When SH_TC = 1 the input SH is interpreted as a signed integer represented in two's complement. The component has a preferred right direction for shifting, which means that when SH > 0 the right shift operation is performed. The input data A is always shifted to the right when SH_TC = 0 (the SH input value is always positive) or when SH_TC = 1 and SH > 0 (the SH input is signed and positive). Otherwise, input A is shifted to the left by $(-2SH_width + SH_{rep})$ bits, where SH_rep is the unsigned integer value of SH.

The arithmetic right shift operation is executed the same way as other arithmetic shifters, the MS bit is copied to all positions that are made open. Differently from other shifters, when shifting to the left, the LS bit is copied to all the positions that are made open. For example, when SH_TC = 1, SH = -2, and $A = (100101)_2$, the output $B = (010111)_2$, where the LS bit of A was replicated on the rightmost positions during the left shift operation. [Table 1-5](#) on page 2 illustrates other combinations of the input values for a small component.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
A	<i>A_width</i>	Input	Input data
SH	<i>SH_width</i>	Input	Shift control
SH_TC	1 bit	Input	Shift two's complement control ■ 0 = Unsigned ■ 1 = Signed
B	<i>A_width</i>	Output	Shifted data out

Table 1-2 Parameter Description

Parameter	Values	Description
A_width	≥ 2	Word length of A and B
SH_width	≥ 1	Word length of SH

Table 1-3 Synthesis Implementations^a

Implementation Name	Function	License Feature Required
mx2	Static implement using 2:1 multiplexers only.	DesignWare
str	Synthesis model targeted for speed	DesignWare
astr	Synthesis model targeted for area	DesignWare

a. During synthesis, Synopsys synthesis tools select the appropriate architecture for your constraints. Alternatively, you may use the set_implementation command to choose one specific implementation from those listed in this table. For details, see the documentation for your Synopsys synthesis tool.

Table 1-4 Simulation Models

Model	Function
dw/dw01/src/DW_sra_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_sra.v	Verilog simulation model source code

The following is a Truth Table for the parameter values $A_width = 8$ and $SH_width = 3$.

Table 1-5 Truth Table ($A_width = 8$, $SH_width = 3$)

SH(2:0)	SH_TC	B(7)	B(6)	B(5)	B(4)	B(3)	B(2)	B(1)	B(0)
000	X	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)
001	X	A(7)	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)
010	X	A(7)	A(7)	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)
011	X	A(7)	A(7)	A(7)	A(7)	A(6)	A(5)	A(4)	A(3)
100	0	A(7)	A(7)	A(7)	A(7)	A(7)	A(6)	A(5)	A(4)
101	0	A(7)	A(7)	A(7)	A(7)	A(7)	A(7)	A(6)	A(5)
110	0	A(7)	A(7)	A(7)	A(7)	A(7)	A(7)	A(7)	A(6)
111	0	A(7)	A(7)	A(7)	A(7)	A(7)	A(7)	A(7)	A(7)

Table 1-5 Truth Table (*A_width* = 8, *SH_width* = 3) (Continued)

SH(2:0)	SH_TC	B(7)	B(6)	B(5)	B(4)	B(3)	B(2)	B(1)	B(0)
100	1	A(3)	A(2)	A(1)	A(0)	A(0)	A(0)	A(0)	A(0)
101	1	A(4)	A(3)	A(2)	A(1)	A(0)	A(0)	A(0)	A(0)
110	1	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)	A(0)	A(0)
111	1	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)	A(0)

Related Topics

- [Logic – Combinational Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp.all;

entity DW_sra_inst is
    generic (
        inst_A_width : POSITIVE := 8;
        inst_SH_width : POSITIVE := 3
    );
    port (
        inst_A : in std_logic_vector(inst_A_width-1 downto 0);
        inst_SH : in std_logic_vector(inst_SH_width-1 downto 0);
        inst_SH_TC : in std_logic;
        B_inst : out std_logic_vector(inst_A_width-1 downto 0)
    );
end DW_sra_inst;

architecture inst of DW_sra_inst is

begin

    -- Instance of DW_sra
    U1 : DW_sra
        generic map ( A_width => inst_A_width, SH_width => inst_SH_width )
        port map ( A => inst_A, SH => inst_SH, SH_TC => inst_SH_TC, B => B_inst );

end inst;

-- pragma translate_off
configuration DW_sra_inst_cfg_inst of DW_sra_inst is
for inst
end for; -- inst
end DW_sra_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_sra_inst( inst_A, inst_SH, inst_SH_TC, B_inst );

parameter A_width = 8;
parameter SH_width = 3;

input [A_width-1 : 0] inst_A;
input [SH_width-1 : 0] inst_SH;
input inst_SH_TC;
output [A_width-1 : 0] B_inst;

    // Instance of DW_sra
    DW_sra #(A_width, SH_width)
        U1 ( .A(inst_A), .SH(inst_SH), .SH_TC(inst_SH_TC), .B(B_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
September 2018	DWBB_201806.2	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 4 ■ Added this Revision History table and the document links on this page

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